# 方鍾豪 Mike Fang

Lead Software Engineer / Team Leader

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## SUMMARY

I'm a Lead Software Engineer with over 10 years of experience across cybersecurity, SaaS, digital advertising, and HR tech. My core strengths lie in full-stack development (TypeScript, Next.js, NestJS, PostgreSQL, AWS), system design, and team leadership. Throughout my career, I've led cross-functional engineering teams and driven end-to-end development of key products—from internal platforms to large-scale web services. I've successfully managed project timelines, recruited and coached engineers, and created a culture of ownership and technical excellence. My leadership approach emphasizes clarity, trust, and long-term code quality. Beyond coding, I focus on building scalable and maintainable systems that align with business goals. Whether optimizing performance, designing architectures, or mentoring engineers, I aim to deliver real-world impact through technology.

## EXPERIENCE

### Lead Software Engineer — Bruce HR Consulting

Taipei, Taiwan · 12/2024 – now

*Sole lead software engineer of BHRC's internal recruitment, sales and market-research CRM. I designed, built and operate the whole platform: backend, frontend, database, AI, cloud and deployment. Used daily by consultants, sales and leadership across tens of thousands of candidate profiles.*

**Product impact**

- Built and run the CRM behind BHRC's recruiting business. It's used daily by consultants, sales and leadership, manages a candidate database of **tens of thousands of profiles**, and covers the whole hiring flow from sourcing to placement.
- Built a **Salary Report** that shows pay against years of experience across the candidate database. Consultants can filter by job function, location, project, job opening and date, remove outliers, and export the chart. They use it to give clients and candidates salary advice backed by data.
- Built a **Sales Sheet** module that tracks consultant targets, deals and replacement hires in one place, replacing spreadsheet-based tracking.

**AI & search features**

- Built **AI candidate–job matching** that finds the **top 100 candidates for a job opening across the entire candidate database** in one click. It compares candidates and the job description on 5 areas (must-have skills, domain experience, functional experience, education, overall profile), combined with hard requirement filters and normalized skill names.
- Kept matching **fast and cheap**: similarity is calculated inside the database (pgvector), and each candidate's text embeddings are reused until their profile changes. Running a match again costs **almost nothing in AI spend**.
- Built an **AI resume-vs-job match review** that gives a match score with reasons for and against, strengths, red flags and suggested improvements. Consultants can then ask the AI follow-up questions about the match.
- Launched an **AI Earning Call Reader** that has turned **175 earnings calls** into hiring leads for consultants in its first two months. It includes a model-evaluation step so AI upgrades are checked before rollout.
- Launched an **AI Data Explorer** that lets non-technical staff ask questions about CRM data in plain English, without filing report requests.
- Built **AI usage and cost dashboards** with per-user breakdowns, so leadership can see and control LLM spending for the first time.

**Performance & engineering speed**

- Made backend builds **6× faster (~12s → ~2s)** by moving to the SWC compiler.
- Made linting **12× faster (~25s → ~2s)** by moving to oxlint, keeping 77 of 84 existing lint rules.
- Cut frontend build time by **~23% (~31s → ~24s)** and type-checking by **~28% (15.8s → 11.3s)** by trimming the bundle.
- Sped up page loads across the main dashboards by running server requests in parallel, caching shared reference data and loading heavy tables and charts only when needed, guided by Lighthouse audits.

**AI-assisted engineering (Claude Code & Cursor)**

- Set up an AI-first engineering workflow in which Claude Code and Cursor follow the same rules from **one shared instruction file**.
- Wrote **36 custom Claude Code skills and 18 Cursor rules** that make agents follow team standards: spec-first design, safe database migrations, security and performance audits, and verification before any change counts as done.
- Used AI agents to run security audits, performance reviews and UI design critiques, then shipped the fixes, all gated by **automated tests and type checks**.
- Set up safe AI-agent access: agents **can't read secrets** and get **least-privilege access** to data.

**Key challenge: redesigning access control without disrupting users**

- **Problem:** An internal review found that access rules were broader than the business intended, so some users could see records outside their responsibilities.
- **Solution:** Redesigned authorization so access follows each user's real working relationship to a record, and applied it consistently across every feature that touches that data.
- **Result:** Fixed it with no downtime, no data migration and no disruption for legitimate users.

### Senior Full Stack Engineer — Placements.io

Taipei, Taiwan · 11/2022 – 10/2024

*Advertise Order Management System*

- Develop feature with React/Redux with TypeScript as frontend and Ruby on Rails as backend, collaborating with U.S. team

### Lead Software Engineer — VYOND

Taipei, Taiwan · 01/2022 – 10/2022

*Online SaaS Animation Software Tool for Businesses*

- Being a project leader to develop new features of our SaaS product with engineers, designers, data specialists, animators and PMs between Taiwan and Hong Kong
- Hire talented engineers into teams and provide coach guideline
- Review the technical specification and provides suggestions to engineers
- Collaborate with machine learning team to provide AI features to users.

### Senior Software Engineer — VYOND

Taipei, Taiwan · 01/2020 – 12/2021

*Online SaaS Animation Software Tool for Businesses*

- Being a project leader to develop new features of our SaaS product with engineers, designers, data specialists, animators and PMs between Taiwan and Hong Kong
- Improve product system stability and performance
- Cultivate new comers to make a good working environment in Taiwan Office
- Design and implement a request scheduler system to handle burst video generation requests request and reduce 60% unnecessary requests
- Lead a new version of video management project to provide better user experience. Including code refactor and unit tests
- Head a content share mechanism project to solve customer pain points

### Software Engineer — VYOND

Taipei, Taiwan · 09/2018 – 12/2019

*Online SaaS Animation Software Tool for Businesses*

- Develop new features with engineers, designers, data specialists, animators between Taiwan and Hong Kong
- Lead a customer issues task force and propose solution to enhance our product
- Reduce 80% unnecessary search results in the search improvement task force
- Increase our retention rate about 40% in the onboarding improvement task force

### Senior Software Automation Engineer — NETGEAR

Taipei, Taiwan · 05/2018 – 07/2018

*Smart Home Solution*

- Maintain and improve existing REST API automation framework. Save 20% time of daily regression
- Payment flow API integration testing and daily regression trouble shooting
- Load/Volume testing for the web server
- Cooperate with colleagues from USA/India/Russia/Taiwan

### Senior Software Engineer — Trend Micro

Taipei, Taiwan · 01/2016 – 05/2018

*Dynamic Sandbox Analysis for Unknown Malware Detection*

- Design and develop the process of sandbox module update to increase deployment flexibility and decrease the response time for detecting a new threat
- Investigate and adopt multiple hypervisors and networking solutions to sandbox environment to improve overall performance
- Design and develop a framework for sandbox result analysis and integrate core techniques modules to improve malware detection rate
- Maintain a Windows application to help customer creates their customized virtual environment
- Maintain CI Jenkins system and write unit test
- Cooperate with the Technical Support team to solve customer feedback issues.
- Act as a communication bridge between developers and QA engineers

### Software Engineer in Test — Trend Micro

Taipei, Taiwan · 09/2013 – 12/2015

*Dynamic Sandbox Analysis for Unknown Malware Detection*

- Have code review/end-to-end testing to find potential defects and drive the architecture design
- Make test malware samples using C/C++, C#...etc. to trigger specific behavior for integration testing
- Build an automated fast build verification testing (BVT) system as a part of CI system to verify basic build quality. Remove unnecessary manual basic checking
- Create a web control system using Python Flask framework to integrate existing testing systems
- Re-design and enhance existing automation framework. Remove limitations of old framework and fulfill requirements from team members and increase test automation rate by at least 20 %
- Coach new comer and lead them to be familiar with our project and culture
- Get top score in 1-month virus training in Trend Micro
- Act as a proactive contact window between Taiwan team and China team

### Part-time Programmer — CAMEO Info Tech

HsinChu, Taiwan · 07/2011 – 04/2013

*Android and iOS Application Development*

- Interactive Picture Book with Corona SDK in Lua language
- An native iOS APP for WIFI storage with WebDAV protocol with Objective-C
- A native Android hot-spot setting widget via NFC

## EDUCATION

### Master of Engineering / Computer Science — National Chiao Tung University

Hsinchu, Taiwan · 09/2011 – 07/2013 · GPA 4.3/4.3

*Thesis: A Decision-theoretic Data Prefetch Algorithm for Reducing Latency and Amortizing Traffic of Mobile Applications.*

### Bachelor of Engineering / Computer Science — National Chiao Tung University

Hsinchu, Taiwan · 09/2007 – 06/2011 · GPA 4.16/4.3

## SKILLS

**General:** Web Application Development, System Software Development, Test Plan and Cases Design, Automation Testing, Regression Testing, Performance Testing Design and Improvement, Continuous Integration and Delivery, Agile and Scrum experience, Certified Scrum Master, Trend Micro Certified Cyber Security Professional, Members mentoring, Communication, Project Planning, Managing and Leading, Full-stack Development, System Design, Access Control, Mobile Application Development, Quick Learning

**Programming Languages:** JavaScript, Python, Shell Script, Java, C/C++, C#, Perl, Ruby, TypeScript, Lua, Objective-C

**Web Developments:** React.js, Redux, Node.js, NestJS, Express.js, Flask, MySQL, Redis, HTML5, CSS3, SCSS, Gatsby.js, Next.js, Restful API, Open API, AVA, Cypress, AWS Services, ELK, Ruby on Rails

**AI:** LLM Application Development, Text Embeddings, Semantic Search, pgvector, Model Evaluation, Claude Code, Cursor

**Tools & Platforms:** Linux, Windows, MacOS, git, Perforce, VSCode, Lighthouse, QEMU, VirtualBox, Docker, Jenkins, Postman, Jmeter, Vyond, Trello, Jira, Selenium, PostgreSQL

**Natural Languages:** English (Business Fluency, TOEIC 940), Chinese (Native Speaker), Japanese (Beginner)